

# IMO 2026 — Problem 1

Solution by Claude Fable 5

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**Source:** `results/imo-2026/claude-fable-5/problem-01/current.md` — autonomous run, no human intervention; solution as written by the model.

## Problem

There are 2026 integers greater than 1 written on a blackboard, not necessarily different. In a move, Confucius chooses two integers  $m > 1$  and  $n > 1$  from different places on the blackboard and replaces these two integers with  $\gcd(m, n)$  and  $\frac{\text{lcm}(m, n)}{\gcd(m, n)}$ . He continues to make moves while it is possible to do so. (a) Prove that, regardless of the choices of Confucius, after finitely many moves, exactly one integer  $M$  on the blackboard is greater than 1. (b) Prove that the value of  $M$  does not depend on the choices of Confucius.

## Solution document

### Status

solved

### Approaches tried

- per-prime-gcd-invariant — worked (round 1, APPROVED): conserved quantity  $g_p = \gcd$  of the  $p$ -adic valuations of the board at every prime  $p$ , plus the lex monovariant  $(S, C)$  ( $S$  = total Omega,  $C$  = count of entries  $> 1$ ) for termination. Yields both parts and the explicit formula  $M = \prod_p p^{\{\gcd(v_p(a_1), \dots, v_p(a_{2026}))\}}$ . Verified adversarially: key algebra re-derived from scratch, 400 random boards  $\times$  3 runs each, strict lex descent at every move, terminal value always matched the formula.
- newman-confluence — worked (round 1, APPROVED): independent second complete proof by abstract rewriting — monovariant  $N = 2027 \cdot P + C$  for termination, prime-support persistence for (a), explicit integer-level local-confluence joins (overlap case via  $\gcd(m, n, k)$  pull-down + pair-collapse, with the join identity  $\gcd(| - |, |\min(, ) - |) = \gcd(| - |, | - |)$ ), and Newman's lemma proved inline by strong induction on  $N$ . Establishes uniqueness of the whole terminal multiset without computing  $M$ . Join identity re-verified on 5000 random triples.

## Current best

Both parts (a) and (b) fully proved (two independent complete proofs). The terminal value is  $M = \prod_p p^{\{g_p\}}$  with  $g_p = \gcd(v_p(a_1), \dots, v_p(a_{2026}))$ .

## Full proof

(Primary proof: approach `per-prime-gcd-invariant`, reviewer-verified round 1. An independent second complete proof via local confluence and Newman's lemma is in `approaches/newman-confluence.md`, also reviewer-verified.)

Throughout, the *board* is the multiset of 2026 integers currently written, in fixed positions (a move replaces the two chosen entries in their places, so the board always has exactly 2026 entries). We write  $v_p(x)$  for the  $p$ -adic valuation of a positive integer  $x$ , i.e. the largest  $e \geq 0$  with  $p^e \mid x$ ; so  $v_p(1) = 0$  for every prime  $p$ , and for each fixed  $x$  we have  $v_p(x) = 0$  for all but finitely many primes  $p$ . We use throughout the **Fundamental Theorem of Arithmetic** (unique prime factorization; `knowledge_base.md`, Number Theory): every positive integer  $x$  satisfies  $x = \prod_p p^{v_p(x)}$ , the product taken over all primes, with all but finitely many exponents zero; consequently, for positive integers  $x, y$ :

- $x \mid y$  if and only if  $v_p(x) \leq v_p(y)$  for every prime  $p$ , and
- $x = y$  if and only if  $v_p(x) = v_p(y)$  for every prime  $p$ .

The overall strategy is the standard **invariant/monovariant method** (`knowledge_base.md`: “Invariants & monovariants”, Combinatorics; “Invariant / monovariant”, General Proof Methods): a strictly decreasing quantity forces termination, and an exactly conserved quantity pins down the terminal state.

### Step 0. The moves are well defined and the board consists of positive integers.

Initially all entries are integers greater than 1, in particular positive. Suppose all entries are positive integers before a move on entries  $m > 1, n > 1$ . The outputs are  $g := \gcd(m, n)$  and  $L/g$  where  $L := \text{lcm}(m, n)$ . We have  $g \geq 1$ . Moreover  $g \mid m$  and  $m \mid L$  (by definition of lcm,  $L$  is a common multiple of  $m$  and  $n$ ), so  $g \mid L$  and  $L/g$  is a positive integer. Hence after the move all entries are again positive integers. By induction on the number of moves, every board arising in the process consists of 2026 positive integers. (Entries equal to 1 can and do appear; the rules only forbid *choosing* an entry equal to 1.)

### Lemma 1 (valuations of gcd and lcm, and of a move).

For positive integers  $m, n$  and any prime  $p$ , with  $a := v_p(m)$ ,  $b := v_p(n)$ :

1.  $v_p(\gcd(m, n)) = \min(a, b)$ ;
2.  $v_p(\text{lcm}(m, n)) = \max(a, b)$ ;
3.  $v_p(\text{lcm}(m, n)/\gcd(m, n)) = \max(a, b) - \min(a, b) = |a - b|$ .

**Proof.** (1) Let  $d := \prod_p p^{\{\min(v_p(m), v_p(n))\}}$  (a finite product, since all but finitely many exponents vanish). By the divisibility criterion above,  $d \mid m$  and  $d \mid n$ , so  $d$  is a common divisor. Conversely, if  $e \mid m$  and  $e \mid n$ , then for every prime  $p$  we have  $v_p(e) \leq v_p(m)$  and  $v_p(e) \leq v_p(n)$ , hence  $v_p(e) \leq \min(v_p(m), v_p(n)) = v_p(d)$ , so  $e \mid d$ . Thus  $d$  is the greatest common divisor:  $\gcd(m, n) = d$ , and  $v_p(\gcd(m, n)) = \min(a, b)$ .

- (2) Symmetrically, let  $l := \prod_p p^{\max(v_p(m), v_p(n))}$ . Then  $m \mid l$  and  $n \mid l$ , so  $l$  is a common multiple; and if  $m \mid f$  and  $n \mid f$  then  $v_p(f) \geq \max(v_p(m), v_p(n)) = v_p(l)$  for all  $p$ , so  $l \mid f$ . Thus  $\text{lcm}(m, n) = l$  and  $v_p(\text{lcm}(m, n)) = \max(a, b)$ .
- (3) Since  $\text{gcd}(m, n) \mid \text{lcm}(m, n)$  (Step 0), the quotient is a positive integer and valuations subtract:  $v_p(\text{lcm}(m, n)/\text{gcd}(m, n)) = v_p(\text{lcm}(m, n)) - v_p(\text{gcd}(m, n)) = \max(a, b) - \min(a, b)$ . Finally  $\max(a, b) - \min(a, b) = |a - b|$ : if  $a \geq b$  this is  $a - b = |a - b|$ , and if  $a < b$  it is  $b - a = |a - b|$ .

So in exponent language: a move on entries with  $p$ -valuations  $(a, b)$  replaces them, at every prime  $p$  simultaneously, by  $(\min(a, b), |a - b|)$ .

**Part (a): termination, and the shape of the terminal board.**

Define, for the current board  $(x_1, \dots, x_{2026})$ :

- $S := \sum_{i=1}^{2026} \Omega(x_i)$ , where  $\Omega(x) := \sum_p v_p(x)$  is the number of prime factors of  $x$  counted with multiplicity (a finite sum;  $\Omega(1) = 0$ , and  $\Omega(x) \geq 1$  for  $x > 1$  since  $x > 1$  has at least one prime factor by the Fundamental Theorem of Arithmetic);
- $C := \#\{i : x_i > 1\}$ .

Both  $S$  and  $C$  are nonnegative integers, and  $C \leq 2026$ .

**Lemma 2 (strict lex descent).** *Every move strictly decreases the pair  $(S, C)$  in the lexicographic order on  $N \times N$ : either  $S$  strictly decreases, or  $S$  stays equal and  $C$  strictly decreases.*

**Proof.** Consider a move on entries  $m > 1$ ,  $n > 1$ , with outputs  $g = \text{gcd}(m, n)$  and  $q = \text{lcm}(m, n)/\text{gcd}(m, n)$ . All other entries are untouched, so the changes in  $S$  and  $C$  come only from replacing the pair  $(m, n)$  by  $(g, q)$ .

*Change in  $S$ .* Fix a prime  $p$  and set  $a := v_p(m)$ ,  $b := v_p(n)$ . By Lemma 1 the  $p$ -part of the touched pair's contribution to  $S$  changes from  $a + b$  to  $\min(a, b) + |a - b|$ . Writing  $a + b = \min(a, b) + \max(a, b)$  and  $|a - b| = \max(a, b) - \min(a, b)$ , the new contribution is  $\min(a, b) + \max(a, b) - \min(a, b) = \max(a, b)$ , so the change at  $p$  is  $\max(a, b) - (\min(a, b) + \max(a, b)) = -\min(a, b)$ . Summing over all primes (a finite sum),

$$\Delta S = -\sum_p \min(v_p(m), v_p(n)) = -\Omega(\text{gcd}(m, n)) \quad (\text{using Lemma 1(1)}).$$

Hence  $\Delta S \leq 0$  always, and  $\Delta S < 0$  if and only if  $\text{gcd}(m, n) > 1$  (since  $\Omega(x) \geq 1$  for  $x > 1$  and  $\Omega(1) = 0$ ).

*Change in  $C$ .* The move removes two entries that are  $> 1$  ( $m$  and  $n$ ) and inserts two entries ( $g$  and  $q$ ), of which at most two are  $> 1$ . Hence  $C$  cannot increase at any move:  $C_{\text{new}} \leq C_{\text{old}}$ .

Now split into two exhaustive, disjoint cases according to  $\text{gcd}(m, n)$ .

**Case 1:  $\text{gcd}(m, n) > 1$**  (this includes the case  $m = n$ , where  $\text{gcd}(m, m) = m > 1$ , with outputs  $(m, 1)$ ). Then  $\Delta S = -\Omega(\text{gcd}(m, n)) \leq -1$ , so  $S$  strictly decreases; since  $S$  is the first lex coordinate,  $(S, C)$  strictly lex-decreases regardless of how  $C$  moves (and  $C$  does not increase in any case, as shown).

**Case 2:  $\text{gcd}(m, n) = 1$ .** Then  $\Delta S = -\Omega(1) = 0$ , so  $S$  is unchanged. The outputs are  $g = 1$  and  $q = \text{lcm}(m, n)/1 = mn > 1$  (as  $m, n > 1$ ). So the two removed entries were both  $> 1$

and exactly one inserted entry is  $> 1$ :  $C$  decreases by exactly 1. Hence  $S$  is equal and  $C$  strictly decreases, again a strict lex decrease.

In both cases  $(S, C)$  strictly decreases lexicographically.

**Lemma 3 (well-foundedness of lex order on  $N \times N$ ).** *There is no infinite strictly lex-decreasing sequence  $(S_0, C_0) > (S_1, C_1) > \dots$  in  $N \times N$ .*

**Proof.** Suppose such an infinite sequence existed. The first coordinates satisfy  $S_0 \geq S_1 \geq \dots$  (in a lex decrease the first coordinate never increases), a non-increasing sequence of nonnegative integers, so it is eventually constant: there is  $K$  with  $S_k = S_K$  for all  $k \geq K$ . For  $k \geq K$ , the lex decreases  $(S_k, C_k) > (S_{k+1}, C_{k+1})$  with equal first coordinates force  $C_K > C_{K+1} > C_{K+2} > \dots$ , an infinite strictly decreasing sequence of nonnegative integers — impossible.

**Termination.** By Lemma 2, the boards after 0, 1, 2, ... moves give a strictly lex-decreasing sequence of pairs in  $N \times N$ ; by Lemma 3 this sequence cannot be infinite. Hence, regardless of choices, only finitely many moves occur: the process reaches a board on which no move is possible.

**Terminal shape (at most one entry  $> 1$ ).** A move is possible exactly when the board contains two entries, in different places, each greater than 1 — the rules impose no other restriction on the choice. (“He continues while possible”: the process stops precisely at a board admitting no move.) Hence the terminal board has at most one entry greater than 1, i.e.  $C \leq 1$ ; by Step 0 all its entries are positive integers, so all entries except possibly one are equal to 1.

It remains to rule out the all-ones terminal board ( $C = 0$ ); we do this with the exact invariant below, which also proves part (b).

### The exact invariant.

**Conventions on gcd of a multiset of nonnegative integers.** Recall that every integer divides 0. For a finite multiset  $T$  of nonnegative integers, if  $T$  contains a positive element, the set of common divisors of  $T$  is a nonempty finite set of positive-or-negative integers bounded by that positive element, and we define  $\gcd(T)$  as its largest element (so  $\gcd(T) \geq 1$ , since 1 is always a common divisor). If all elements of  $T$  are 0, every positive integer is a common divisor and we set  $\gcd(T) := 0$ . With these conventions,  $\gcd(T) = 0$  if and only if all elements of  $T$  are 0; and in all cases the set of common divisors of  $T$  is exactly the set of divisors of  $\gcd(T)$  — for the all-zero multiset because every integer divides  $0 = \gcd(T)$ , and otherwise by the classical characterization proved as follows: if  $d = \gcd(T) > 0$  and  $e$  is a common divisor of  $T$ , then so is  $\text{lcm}(d, e)$  — indeed each  $t$  in  $T$  is a common multiple of  $d$  and  $e$ , hence a multiple of  $\text{lcm}(d, e)$  (by the defining minimality of  $\text{lcm}$  as established in Lemma 1(2), every common multiple of  $d$  and  $e$  is divisible by  $\text{lcm}(d, e)$ ) — so  $\text{lcm}(d, e) \leq d$  by maximality of  $d$ , while  $d \mid \text{lcm}(d, e)$  forces  $\text{lcm}(d, e) = d$ , i.e.  $e \mid d$ . Conversely any divisor of  $d$  divides every element of  $T$ .

**Definition.** For a prime  $p$  and a board  $B = (x_1, \dots, x_{2026})$ , let

$g_p(B) := \gcd\{v_p(x_1), \dots, v_p(x_{2026})\}$  (multiset gcd with the conventions above).

**Lemma 4 (Euclidean subtraction identity, all cases).** *For nonnegative integers  $a, b$ , the pairs  $\{a, b\}$  and  $\{\min(a, b), |a - b|\}$  have the same set of common divisors; in particular  $\gcd(a, b) = \gcd(\min(a, b), |a - b|)$ .*

**Proof.** By symmetry assume  $a \geq b$ , so  $\min(a, b) = b$  and  $|a - b| = a - b$ . If  $d \mid a$  and  $d \mid b$ , then  $d \mid (a - b)$  (a difference of multiples of  $d$ ) and  $d \mid b$ . Conversely, if  $d \mid b$  and  $d \mid (a - b)$ , then  $d \mid (b + (a - b)) = a$  and  $d \mid b$ . So the common-divisor sets coincide. This argument uses nothing about positivity and is valid in all four exponent cases:  $a = b > 0$  (then the second pair is  $(a, 0)$ , and  $d \mid a, d \mid 0$  iff  $d \mid a, d \mid a$ );  $a \neq b$  both positive; exactly one of  $a, b$  zero (say  $b = 0$ : the second pair is  $(0, a)$ , same divisor conditions); and  $a = b = 0$  (both pairs are  $(0, 0)$ ). Equal common-divisor sets give equal gcds under our conventions (both the all-zero case, where both gcds are 0, and the case of a positive element, where both gcds are the largest common divisor).

**Lemma 5 (one-move invariance of  $g_p$ ).** *For every prime  $p$ , a single move does not change  $g_p$  of the board.*

**Proof.** Fix  $p$ . The move touches two positions, changing their valuations from  $(a, b)$  to  $(\min(a, b), |a - b|)$  by Lemma 1; the other 2024 positions are untouched. Let  $R$  be the multiset of valuations of the untouched positions. An integer  $d$  is a common divisor of the multiset  $R \cup \{a, b\}$  if and only if  $d$  divides every element of  $R$  and  $d$  is a common divisor of  $\{a, b\}$ ; by Lemma 4 the latter is equivalent to  $d$  being a common divisor of  $\{\min(a, b), |a - b|\}$ . Hence  $R \cup \{a, b\}$  and  $R \cup \{\min(a, b), |a - b|\}$  have the same set of common divisors, and therefore the same gcd (same reasoning as at the end of Lemma 4: in the all-zero case both gcds are 0 — note the two multisets are all-zero simultaneously, since  $a = b = 0$  iff  $\min(a, b) = |a - b| = 0$  — and otherwise both gcds are the largest element of the common set of common divisors). So  $g_p(\text{board})$  is unchanged.

**Corollary (invariance along the whole process).** *Let  $B_0$  be the initial board and  $B_k$  the board after  $k$  moves (any choices). Then for every prime  $p$  and every  $k \geq 0$ ,  $g_p(B_k) = g_p(B_0)$ .*

**Proof.** Induction on  $k$ . Base  $k = 0$ : trivial. Step: assume  $g_p(B_k) = g_p(B_0)$ ; the  $(k+1)$ -st move turns  $B_k$  into  $B_{k+1}$ , and Lemma 5 gives  $g_p(B_{k+1}) = g_p(B_k) = g_p(B_0)$ .

**Part (a), completed: exactly one entry  $> 1$  remains.**

We showed the process terminates after finitely many moves at a board  $B^*$  with at most one entry greater than 1. Suppose, for contradiction, that all entries of  $B^*$  equal 1. The initial board contains  $a_1 > 1$ , which by the Fundamental Theorem of Arithmetic has a prime factor  $p$ , so  $v_p(a_1) \geq 1$ . Thus the initial valuation multiset at  $p$  contains a positive element, so  $g_p(B_0) \geq 1$  by our conventions. But if  $B^*$  is all ones, every valuation on  $B^*$  is  $v_p(1) = 0$ , so  $g_p(B) = \gcd(0, \dots, 0) = 0$ . This contradicts the Corollary, which gives  $g_p(B) = g_p(B_0) \geq 1$ . Hence  $B^*$  is not all ones, so it has exactly one entry  $M > 1$ , and the other 2025 entries equal 1. This proves part (a).

**Part (b):  $M$  is independent of the choices.**

Let  $B_0 = (a_1, \dots, a_{2026})$  be the initial board, and let  $B^*$  be any terminal board reached by any sequence of choices; by part (a),  $B^*$  consists of one entry  $M > 1$  and 2025 entries equal to 1. Fix a prime  $p$ . The valuation multiset of  $B^*$  at  $p$  is  $\{v_p(M), 0, 0, \dots, 0\}$  (2025 zeros). Its gcd is  $v_p(M)$ : if  $v_p(M) = 0$  this is the all-zero multiset with  $\gcd 0 = v_p(M)$ ; if  $v_p(M) > 0$ , the common divisors of  $\{v_p(M), 0, \dots, 0\}$  are exactly the divisors of  $v_p(M)$  (every integer divides 0), the largest of which is  $v_p(M)$ . Hence  $g_p(B^*) = v_p(M)$ , and by the Corollary,

$v_p(M) = g_p(B^*) = g_p(B_0) = \gcd(v_p(a_1), \dots, v_p(a_{2026}))$  for every prime  $p$ .

Only finitely many primes divide  $a_1 \cdot \dots \cdot a_{2026}$ , and for any other prime  $p$  all  $v_p(a_i) = 0$ , so  $g_p(B_0) = 0$ ; thus all but finitely many exponents below vanish and the product is finite. By the Fundamental Theorem of Arithmetic, a positive integer is determined by its valuations:

$$M = \prod_p p^{v_p(M)} = \prod_p p^{\gcd(v_p(a_1), \dots, v_p(a_{2026}))}.$$

The right-hand side depends only on the initial board  $B_0$ , not on any of the choices made. Therefore every sequence of choices terminates with the same integer  $M$ . This proves part (b).